Rings and modular arithmetic exercises

1. Find all units and nonzero zero divisors in $\mathbb{Z}_8$, and compute the inverse of 3.
2. Fix an integer a. Determine whether polynomial evaluation $ev_a : \mathbb{Z}[x] \to \mathbb{Z}_{12}, p(x) \mapsto [p(a)]_{12}$, is a ring homomorphism. Find its entire kernel and image. What changes if the codomain is $\mathbb{Z}$? Give a quotient-ring isomorphism from the first isomorphism theorem in each case.
3. Use the extended Euclidean algorithm to solve $17x \equiv 5 \pmod{43}$.
4. Solve $x \equiv 1 \pmod{4}, x \equiv 2 \pmod{5}, x \equiv 3 \pmod{7}$, and find the least nonnegative solution.
5. Let $I = \{2k : k \in \mathbb{Z}\}$ and $J = \{[0], [2], [4]\}$. Decide whether they are ideals in their respective rings $\mathbb{Z}$ and $\mathbb{Z}_6$. Determine the structure of $\mathbb{Z}_6/J$.
6. Prove that if R is an integral domain and $a, b \neq 0$, then $ab \neq 0$. Give a counterexample in a ring that is not an integral domain.
7. Solve $10x \equiv 14 \pmod{26}$ completely, listing all least nonnegative solutions. Then solve $x \equiv 3 \pmod{10}, x \equiv 8 \pmod{15}$. If solutions exist, identify the correct modulus for uniqueness; otherwise disprove compatibility. Finally replace the second residue 8 by 9 and decide again.
8. Find all general ring homomorphisms $\varphi : \mathbb{Z}_4 \to \mathbb{Z}_6$, without assuming preservation of 1. First prove that $e = \varphi([1]_4)$ determines the map, and rigorously derive the necessary conditions on e. Find every kernel and image and identify any unital maps. Testing a single formula is not a complete classification.
9. In $M_2(\mathbb{Z})$, let $S = \{\mathrm{diag}(a, 0) : a \in \mathbb{Z}\}$. Prove that it is a subring, find its own multiplicative identity, and decide whether it is a unital subring or ideal of the ambient ring. Find the inverses of

$$P = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}, \qquad Q = \begin{pmatrix} 1 & 2 \\ 2 & 6 \end{pmatrix}$$

in $M_2(\mathbb{Q})$. Which are also units of $M_2(\mathbb{Z})$? Explain the integer restriction on the determinant.

Rings and modular arithmetic exercises (continued)

10. On $\mathbb{Q}$, define $a \oplus b = a + b - 5$ and $a \otimes b = ab - 5a - 5b + 30$. Instead of expanding every axiom, construct an explicit ring isomorphism to ordinary $\mathbb{Q}$. Find the new zero, multiplicative identity, additive inverse of arbitrary a, invertible elements and their inverses. Compute and check the inverse of 7.
11. Construct an isomorphism $\mathbb{Z}_{18} \cong \mathbb{Z}_2 \times \mathbb{Z}_9$ and its inverse. Find every idempotent $x^2 = x$, prove the list complete, and identify the units and nonzero zero divisors among them. Do not merely test all 18 values. Also count and list all units, relating the result to the gcd criterion.

References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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https://github.com/xinhi6716/computer-science-self-study

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